

Cost and Time Estimation

Sensor Data Collector (Mobile App)

Denis Schatzmann

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1 Task

The Sensor Data Collector lets a user pick sensors from a select box, add them to a list (duplicates ignored), remove a marked sensor, and toggle data collection with a Start/Stop button. On stop the collected data is sent to a backend service. This report follows the steps from the lecture: estimate the size, derive effort and schedule with COCOMO, and finally the cost.

2 Size

This is a small app that is mostly user interface with very little logic in the background. For that case the lecture recommends the *widget point* method, so we use it as the main estimate and the function point count as a cross-check.

We count the interface elements:

Widget type	Count
Input widgets (select box, add, remove, Start/Stop)	4
Composite widgets (sensor list)	1
Describing widgets (labels)	3
Widget points	8

Table 1: Widget point count.

With $FP = 2 \times \text{widget points}$ this gives **16 function points**. A classic function point count over the same six functions (the three buttons as external inputs, the backend send as an external output, the select box as an external inquiry, and the sensor list as an internal file) comes out somewhat higher, at about 23 FP. Since the app is essentially a thin user interface over an existing backend, we take the lower widget-point value of 16 FP as the estimate and treat the 23 FP as a conservative upper bound.

3 Effort: COCOMO

Converting to lines of code with the language table (53 LOC/FP for Java/Kotlin) gives $16 \times 53 \approx 850$ LOC, i.e. about 0.85 KDL. Applying COCOMO in organic mode results in a base effort of roughly **2.7 person months**.

Because this is medical software, the required reliability is rated high, which adds about 15 %. The adjusted effort is therefore about **3.1 person months**. Split over the usual phases of a small project:

Phase	Share	Person months
Design	19%	0.6
Implementation	63%	2.0
Integration and tests	18%	0.6

Table 2: Effort distribution for a small project.

4 Schedule

Using the organic schedule formula, 3.1 person months stretch over about **3.8 months** with an average staffing of roughly one person. So a single developer working a bit under four months covers the work.

5 Cost

With a monthly salary of CHF 10'000, the personnel cost is

$$3.1 \times 10\,000 \approx \text{CHF } 31\,000.$$

Adding about CHF 2'000 for tooling licences, test devices and the cloud backend gives a total of about **CHF 33'000** for one developer over roughly **3.8 months**.

6 Accuracy

At this early stage the estimate is only accurate within a factor of about 0.5 to 2, so the realistic range is roughly CHF 17'000 to CHF 66'000. The estimate assumes one cross-platform codebase, an existing backend API, and standard sensor access without custom drivers.